CHAPTER - 07 UNIT - 4

MEASURES OF CENTRAL TENDENCY

Def: A set of numerical facts collected is called a set a data.

Variable: A quantity whose value can vary from one individual to another is called variable.

Ex: marks, heights, weight, age etc

Continous variable: A variable which can take any numerical value within a certain range is called a continous variable.

Ex: Interval [2,3], [3,4] ---- etc

Discrete variable: A variable which is incapable of taking all possible numerical values is called a discrete variable.

Measures of central tendency: The following are the main measures of central tendency that are in common use: ① mean ② median ③ mode

Mean: mean of any set of observations items is calculated by adding the values of all the observation in the series and dividing by the total number of observation

$$\text{mean} = \frac{\text{Sum of observation}}{\text{Total number of observation}}$$

If x is the variable which takes value $x_1, x_2, x_3 \ldots x_n$ over in items

$$\text{means} = \bar{x} = \frac{x_1 + x_2 + x_3 \ldots + x_n}{x}$$

$$= \frac{\sum_{i=1}^{n} x_i}{n}$$

NOTE: mean, arithmetic means and average means are the Same.

1) Find the means of the data: 3, 12, 5, 20, 10

$$\text{mean} = \frac{x_1 + x_2 + x_3 + x_4 + x_5}{5}$$

$$= \frac{3 + 12 + 5 + 20 + 10}{5}$$

$$= \frac{50}{5} = 10$$

2) The weight (in kilogram) of 6 persons are 67, 65, 64, 71, 45 find the arithmetic means of their weight.

$$\text{mean} = \frac{x_1 + x_2 + x_3 + x_4 + x_5 + x_6}{6}$$

$$= \frac{61+65+63+64+71+74}{6}$$

$$= \frac{378}{6} = 63\,kg$$

3) Find the mean of 7, 6, 10, 0, 9

$$\text{mean} = \frac{x_1 + x_2 + x_3 + x_4 + x_5}{5}$$

$$= \frac{7+6+10+0+9}{5}$$

$$= \frac{32}{5} = 6.4$$

Median : median is defined as the middle value of the series when arranged either in ascending (or) descending order.

Median for ungrouped data:

① If n is odd, medians the value of $\left(\frac{n+1}{2}\right)^{th}$ term.

② If n of even,

median = The value of $\left(\frac{n}{2}\right)^{th}$ term + $\left(\frac{n}{2}+1\right)^{th}$ term

1) Find the median of 7, 9, 4, 3 and 10

On arranging the given term in ascending order

3, 4, 7, 9, 10

Here $n = 5$

median = The value of $\left(\frac{n+1}{2}\right)^{th}$ term

= The value of $\left(\frac{5+1}{2}\right)^{th}$ term

= The value of 3rd term

= middle term

= 7

median = 7

2) Find the median of 20, 14, 15, 12, 2, 8, 6, 7

arrange an ascending order

2, 6, 7, 8, 12, 14, 15, 20

$n = 8$

median = The value of $\left(\frac{n}{2}+1\right)^{th}$ term

= The value of $\left(\frac{8}{2}+1\right)^{th}$ term

= The value 5th term

= middle term

= 12

In median

$$= \frac{\left(\frac{n}{2}\right) + \left(\frac{n}{2}+1\right)}{2} = \frac{8+12}{2} = \frac{20}{2} = 10$$

$$= \frac{4^{th} + 5^{th}}{2}$$

MODE: Mode is the value which occurs (or) repeats itself the greatest number of times (or) that has the highest frequency in the given data.

1) The find mode of the set of data

5, 2, 4, 6, 7, 5, 2, 5, 7, 8, 5, 7, 2

The value 5 occurs 4 times

∴ Mode = 5

2) Find the mode of following distribution

Height (cm)	No. of plants
100	2
150	6
180	7
190	14
250	8
260	9
350	3

Sol: Since the frequency of 190 is the highest it is the Mode = 190

NOTE:

Relation between mean, median, mode

* 3 median = mode + 2 mean

* mode = 3 median − 2 mean.

1) Find the mode, if median is 125.8, mean = 128

mode = 3 median − 2 mean

$= 3 \times 125.8 - 2 \times 128$

$= 377.4 - 256$

$= 121.4$

2) In a unimodal distribution arithmetic mean = 74.5 and median = 76.9 then find the mode

mode $= 3 \times 76.9 - 2 \times 74.5$

$= 230.7 - 149$

$= 81.7$